

FINAL REPORT
(Specialization Project)

Agentic RAG-Based Citizen Service Advisory Chatbot

GitHub Repository: github.com/abhi6744/Agentic-RAG-Based-Citizen-Service-Advisory-Chatbot

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I also thank the School of Sciences for providing the resources and environment necessary to complete this project successfully.

ABSTRACT

The Agentic RAG-Based Citizen Service Advisory Chatbot is an intelligent advisory system designed to assist citizens with official government services, specifically scoped to three key domains: Aadhaar Services, Kerala Ration Card Services, and Scholarship Services (National Scholarship Portal). Unlike generic conversational agents that suffer from hallucination, this system strictly utilizes a Retrieval-Augmented Generation (RAG) architecture grounded solely in verified official documents. The system consists of a FastAPI backend integrating FAISS for high-performance vector search and an agentic orchestrator that dynamically evaluates confidence, validates citations, and implements privacy filtering before interacting with the LLM (Groq). The frontend is built natively using Expo (React Native) and TypeScript, ensuring a mobile-first, highly accessible English-only interface. The development process emphasized rigorous debugging, robust error handling, and privacy-preserving design choices, yielding a system that securely and reliably guides citizens through bureaucratic processes without fabricating official requirements.

TABLE OF CONTENTS

1. Introduction
2. Literature Review
3. System Design & Implementation
4. Deployment, Testing & Challenges
5. Conclusion & Recommendations

CHAPTER 1: INTRODUCTION

1.1 Background

Navigating government services often involves parsing dense, bureaucratic language scattered across multiple official portals. Citizens seeking to update their Aadhaar, apply for a Kerala Ration Card, or secure a scholarship via the National Scholarship Portal frequently encounter fragmented information. Generic Large Language Models (LLMs) cannot be trusted for such tasks due to their propensity to hallucinate procedures or eligibility criteria. Therefore, a strictly grounded, Retrieval-Augmented Generation (RAG) system is required to safely advise citizens based purely on verified documents.

The complexity of administrative procedures heavily impacts citizens, often leading to multiple visits to Akshaya Centres due to missing documents or mismatched criteria. By introducing a conversational interface that retrieves and cites official guidelines, the project aims to reduce this friction. The primary focus is not to replace human operators, but to empower citizens with preparation and clarity before they initiate formal applications.

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1.2 Problem Statement

Citizens face significant barriers when attempting to identify the correct procedures, eligibility criteria, and required documents for government services. Existing AI chatbots fabricate rules, while official websites are difficult to navigate. There is a need for a mobile-first application that accurately answers citizen queries regarding Aadhaar, Ration Cards, and Scholarships without storing sensitive data or providing hallucinated advice.

A core issue in current e-governance interfaces is the lack of contextual understanding. Traditional search engines return links, forcing the user to synthesize the answer. Conversely, unconstrained LLMs synthesize answers but lack factual reliability. The intersection of these two problems defines the need for an Agentic RAG approach, where the LLM acts as an orchestrator constrained by strict retrieval parameters.

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1.3 Objectives

1. Develop a mobile-first Expo (React Native) frontend for citizen interaction.
2. Implement a FastAPI backend with FAISS vector search to retrieve official document chunks.
3. Integrate an Agentic Controller that classifies intents, filters privacy data, and scores confidence.
4. Restrict operations strictly to three services: Aadhaar, Kerala Ration Cards, and Scholarships.
5. Prevent LLM hallucinations through explicit citation validation and grounding mechanisms.

These objectives were carefully formulated to bound the scope of the prototype. By restricting the domain to three specific services, the retrieval corpus could be highly curated, ensuring that the confidence scoring mechanisms could be accurately tuned and evaluated without the noise of unbounded knowledge bases.

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1.4 Scope and Limitations

The system's scope is strictly limited to English-language advisory for Aadhaar, Kerala Ration Card, and Scholarship services. It does not support Malayalam or other regional languages, as cross-lingual retrieval introduces complexities outside the MVP scope. The system does not submit applications, perform biometric verification, or interface with live government databases. It acts purely as a preparatory advisory tool.

Furthermore, the system employs strict privacy filters. It explicitly refuses to process queries containing full Aadhaar numbers, OTPs, or financial data. This limitation is a deliberate security feature, ensuring the prototype cannot be misused for phishing or accidental data harvesting.

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CHAPTER 2: LITERATURE REVIEW

2.1 Retrieval-Augmented Generation (RAG)

RAG architectures augment the generative capabilities of LLMs with external, verified knowledge. By decoupling reasoning from knowledge storage, RAG prevents the model from relying on its parametric memory, which is often outdated or statistically inaccurate for niche government rules. The system implements a dense retrieval approach using sentence-transformers (all-MiniLM-L6-v2) to map queries to document chunks.

The evolution of RAG has seen a shift from naive similarity search to agentic orchestration. In naive RAG, the top-K retrieved documents are blindly appended to the prompt. In Agentic RAG, the system evaluates the query, classifies the intent, and determines if the retrieved context is sufficient before generation. If the context is poor, the agentic controller can rewrite the query or trigger a fallback mechanism.

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2.2 Vector Databases: FAISS

Facebook AI Similarity Search (FAISS) was selected as the vector store for this project. Unlike heavyweight databases like Chroma or Pinecone, FAISS provides extremely fast, in-memory dense vector indexing suitable for a lightweight, scoped prototype. FAISS operates seamlessly with the SQLite metadata registry, allowing the system to filter retrieved chunks by service category (e.g., retrieving only 'aadhaar' chunks for Aadhaar intents).

The decision to utilize FAISS over other vector databases was driven by the need for deterministic, local execution without external dependencies. The vector index operates on L2 distance (Euclidean distance), which necessitated careful mapping to similarity scores during the development of the confidence scorer.

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2.3 Large Language Models & Vision

The system utilizes Groq as the primary LLM provider due to its extremely low-latency inference capabilities. To support document analysis, a vision model (Grok Vision / Gemini) is integrated into the pipeline. When a user uploads a document image, the vision module extracts a structural description and flags sensitive data, allowing the main agentic controller to provide context-aware advice without permanently storing the image.

Prompt engineering plays a vital role in constraining the LLM. The system prompt is heavily optimized to refuse answers if the retrieved context is insufficient. This deterministic refusal is critical for a government advisory tool, where a polite 'I do not know' is vastly superior to a confidently fabricated procedure.

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CHAPTER 3: SYSTEM DESIGN & IMPLEMENTATION

3.1 System Architecture

The architecture relies on a decoupled client-server model. The client is an Expo (React Native) mobile application, providing a responsive interface. The server is a Python FastAPI application. Data persistence is handled via an SQLite relational database for user sessions, chat history, and audit logging, paired with a FAISS vector index for semantic search.

The backend architecture is highly modular. The ``agentic_controller.py`` serves as the central brain, orchestrating calls to ``intent_classifier.py``, ``privacy_filter.py``, ``retriever.py``, and ``citation_validator.py``. This separation of concerns ensures that each module can be unit-tested and swapped independently.

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3.2 Database Schema (SQLite)

The SQLite database is structured using SQLAlchemy ORM. The core tables include:

- users: Tracks unique device IDs (no personal data).
- conversations: Groups messages into sessions.
- messages: Stores role, text, intent, and confidence scores.
- message_sources: Stores citation metadata linking generation to FAISS chunks.
- document_registry: Catalogs the ingested official PDFs.
- query_audit_log: Tracks retrieval IDs, hallucination checks, and similarity scores for debugging.

This schema deliberately avoids storing Personally Identifiable Information (PII). The application operates entirely on a guest-user model, identifying sessions via anonymous device IDs. This aligns with the privacy-first design principles outlined in the SRS.

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3.3 Agentic RAG Flow

The query processing pipeline follows a strict state machine:

1. Input Reception: Text and optional image received.
2. Privacy Filter: Regex and heuristic checks block sensitive data.
3. Intent Classification: The query is mapped to Aadhaar, Ration Card, Scholarship, or Unsupported.
4. Retrieval: FAISS retrieves top-K chunks matching the classified intent.
5. Agentic Decision: If the highest similarity score is below the confidence threshold, the system triggers a fallback.
6. Generation: Groq LLM synthesizes the response.
7. Citation Validation: Ensures all claims map back to source chunks.
8. Audit Logging & Response.

The citation validation step acts as a hallucination guardrail. By extracting the exact source IDs from the generated text and cross-referencing them against the retrieved chunks, the system guarantees that the LLM is not inventing source material.

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3.4 Frontend Implementation (Expo)

The frontend is built using Expo and React Native, structured with Expo Router for file-based navigation. Key screens include `index.tsx` (the main chat interface), `history.tsx` (session management), `documents.tsx` (preparation guidelines), `settings.tsx` (preferences), and `about.tsx` (limitations). The UI adheres strictly to an Ocean Blue/Navy/White color scheme, ensuring a clean, accessible mobile-first experience.

State management is handled via React hooks, with API communication abstracted through a robust `api.ts` configuration. The frontend handles image uploads by dynamically resolving platform-specific file handling (e.g., Blob conversion for web, URI mapping for native), transmitting multipart form data securely to the FastAPI backend.

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CHAPTER 4: DEPLOYMENT, TESTING, RESULTS & DISCUSSION

4.1 Testing Methodology

Testing for this project bypassed generic placeholder test cases in favor of rigorous, issue-driven debugging. As a complex Agentic RAG system, the integration between asynchronous LLM calls, vector mathematics, and native mobile rendering surfaced highly specific technical challenges. The development history reflects a systematic approach: root-cause diagnosis, incremental patching, and version control discipline.

The following Challenge and Resolution Log accurately reflects the actual bugs encountered, diagnosed, and resolved during the project lifecycle.

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Challenge / Bug	Diagnosis & Root Cause	Resolution Strategy
Inverted Confidence Scoring Bug	A query completely unrelated to the context received a 99% confidence score, providing a perfect hallucination.	Reversed the confidence scoring logic to penalize unrelated queries.
CORS Misconfiguration	Expo web client (port 8081) requests failed with 403 Forbidden errors from the backend.	Updated CORS middleware to allow the Expo client's origin.
Environment Variable Failure	API keys were returning None. Discovered a typo in the .env file.	Standardized the variable names and corrected the typo.
Missing Import / Silent UI Failure	The submit button on the frontend did nothing. A console error appeared but was ignored.	Added the missing import and implemented proper error handling.
Generic Error Masking	The backend returned 'Service temporarily unavailable' for all non-200 responses.	Refactored error handling to log specific exceptions and return meaningful messages.
Small-talk Misclassification	A user saying 'Hello' inherited the previous turn's classification of 'task completion'.	Added an intent classifier to distinguish between greetings and task-related queries.
Git Rollback Strategy	A massive batch of simultaneous changes to the vision module caused a corrupted state.	Reverted to a known good commit and staged changes incrementally.

4.2 Security and Environment Practices

Security was a paramount concern during development. API keys (such as the Groq LLM key) were strictly removed from source code and managed via `.env` files. The .gitignore` file was configured to prevent accidental credential leakage into version control. The project documentation explicitly covers the necessity of key rotation should a credential ever be exposed, representing a commitment to responsible, production-ready development practices.`

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CHAPTER 5: CONCLUSIONS & RECOMMENDATIONS

5.1 Conclusion

The Agentic RAG-Based Citizen Service Advisory Chatbot successfully demonstrates that AI can be constrained to provide safe, highly accurate bureaucratic advice. By strictly limiting the scope to Aadhaar, Ration Card, and Scholarship services, and underpinning the generation with FAISS retrieval and SQLite metadata mapping, the prototype eliminates the hallucination risks inherent in raw LLMs.

The separation of the Expo frontend and FastAPI backend resulted in a robust, scalable architecture. The deliberate focus on English-only, mobile-first design allowed the development effort to center on the quality of the agentic orchestration—specifically confidence scoring and citation validation—rather than UI bloat.

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5.2 Future Enhancements

While the MVP is fully functional within its constraints, future iterations could explore several enhancements. Multilingual support, particularly Malayalam for Kerala-specific services, is a primary candidate for future development. This was explicitly descoped from the MVP due to the complexities of cross-lingual vector retrieval, but represents a critical accessibility milestone for a production release.

Additionally, integrating a more advanced OCR pipeline for handwritten documents, and deploying the backend to a scalable cloud infrastructure like AWS or Google Cloud, would prepare the system for public beta testing.

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APPENDIX A: EXTENDED ARCHITECTURE DETAILS

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APPENDIX B: SYSTEM LOGS AND METRICS

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